

## LEWISTOWN 42 WSW, MT, USA

WMO: 742280

Lat: 46.8847N

Lon: 110.2895W

Elev: 5070

StdP: 12.20

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 04140

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | -16.4      | -10.6 | -21.1                         | 2.1 | -14.5 | -15.8 | 2.8 | -9.9 | 24.5                  | 40.4 | 21.7 | 39.9 | 2.4                      | N/A  | 0.615 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 30.3                   | 85.9            | 56.8 | 83.2 | 56.0 | 80.2 | 55.6 | 60.5                | 77.6 | 58.9 | 76.8 | 57.4 | 75.8 | 7.7                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 54.0                            | 75.1 | 66.6 | 51.9 | 69.2 | 65.0 | 50.0 | 64.6 | 63.4 | 29.2          | 77.5 | 27.9 | 76.6 | 26.8 | 75.7 | 66.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |
| 20.6              | 17.7 | 15.3 | DB                         | -25.8 | 90.9               | 7.1 | 2.3                                                | -30.9 | 92.6       | -35.0 | 93.9       | -39.0 | 95.2       | -44.2 | 96.9 |
|                   |      |      | WB                         | -26.1 | 64.7               | 7.6 | 1.5                                                | -31.6 | 65.8       | -36.0 | 66.6       | -40.3 | 67.5       | -45.8 | 68.5 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 40.8   | 25.9  | 21.5  | 32.3  | 37.8  | 45.7  | 54.4  | 62.2  | 60.8  | 52.6  | 40.9  | 31.8  | 22.5  |  |  |
|                                                                        | DBStd       | 17.34  | 12.95 | 15.33 | 11.82 | 8.54  | 7.34  | 6.40  | 5.43  | 6.06  | 8.39  | 10.45 | 12.85 | 14.41 |  |  |
|                                                                        | HDD50       | 4452   | 749   | 797   | 550   | 375   | 175   | 27    | 1     | 5     | 66    | 306   | 549   | 852   |  |  |
|                                                                        | HDD65       | 8889   | 1214  | 1217  | 1014  | 816   | 598   | 322   | 118   | 154   | 376   | 746   | 997   | 1317  |  |  |
|                                                                        | CDD50       | 1104   | 1     | 1     | 1     | 9     | 42    | 159   | 380   | 339   | 145   | 25    | 2     | 0     |  |  |
|                                                                        | CDD65       | 63     | 0     | 0     | 0     | 0     | 0     | 4     | 32    | 22    | 5     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 2468   | 0     | 0     | 0     | 2     | 17    | 204   | 1017  | 892   | 315   | 21    | 0     | 0     |  |  |
|                                                                        | CDH80       | 641    | 0     | 0     | 0     | 0     | 0     | 46    | 275   | 255   | 62    | 3     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 5.7    | 7.4   | 6.3   | 6.4   | 5.5   | 4.7   | 5.0   | 4.4   | 4.3   | 4.6   | 5.8   | 7.0   | 7.0   |  |  |
|                                                                        | PrecAvg     | 17.2   | 0.7   | 0.6   | 0.9   | 1.8   | 2.6   | 3.3   | 1.7   | 1.4   | 1.2   | 1.1   | 0.9   | 0.8   |  |  |
| (15) Precipitation                                                     | PrecMax     | 24.7   | 1.2   | 1.4   | 1.7   | 2.9   | 4.6   | 5.7   | 5.4   | 3.6   | 1.9   | 2.5   | 1.8   | 2.1   |  |  |
|                                                                        | PrecMin     | 13.3   | 0.3   | 0.3   | 0.5   | 1.1   | 0.7   | 1.3   | 0.5   | 0.2   | 0.2   | 0.4   | 0.5   | 0.4   |  |  |
|                                                                        | PrecStd     | 3.0    | 0.3   | 0.3   | 0.4   | 0.6   | 0.9   | 1.1   | 1.1   | 0.9   | 0.4   | 0.5   | 0.4   | 0.4   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 54.9  | 56.8  | 62.5  | 71.4  | 76.7  | 85.6  | 89.2  | 90.4  | 85.1  | 77.1  | 65.1  | 51.3  |  |  |
|                                                                        |             | MCWB   | 39.5  | 40.0  | 43.7  | 48.5  | 52.3  | 59.7  | 57.7  | 56.6  | 54.4  | 51.5  | 44.7  | 37.1  |  |  |
|                                                                        | 2%          | DB     | 49.0  | 50.2  | 57.2  | 65.1  | 71.4  | 80.0  | 86.2  | 85.9  | 81.9  | 68.3  | 58.0  | 47.9  |  |  |
|                                                                        |             | MCWB   | 36.3  | 37.5  | 41.4  | 46.4  | 51.5  | 57.5  | 57.3  | 56.2  | 53.5  | 47.7  | 42.4  | 36.3  |  |  |
|                                                                        | 5%          | DB     | 45.9  | 46.1  | 53.4  | 59.4  | 67.7  | 75.0  | 83.3  | 82.8  | 77.7  | 63.6  | 52.6  | 44.3  |  |  |
|                                                                        |             | MCWB   | 35.1  | 35.8  | 39.9  | 43.0  | 50.5  | 56.1  | 57.0  | 55.5  | 52.4  | 46.0  | 39.7  | 34.6  |  |  |
|                                                                        | 10%         | DB     | 42.6  | 41.8  | 49.3  | 54.0  | 63.3  | 70.5  | 80.1  | 79.4  | 73.4  | 58.6  | 48.5  | 40.5  |  |  |
|                                                                        |             | MCWB   | 33.8  | 33.4  | 37.6  | 40.7  | 48.0  | 53.8  | 56.5  | 55.1  | 51.3  | 44.0  | 37.7  | 32.3  |  |  |
|                                                                        | 0.4%        | WB     | 40.0  | 41.1  | 45.5  | 50.4  | 55.7  | 62.3  | 63.9  | 60.2  | 59.2  | 53.2  | 46.2  | 39.4  |  |  |
|                                                                        |             | MCDB   | 52.5  | 53.8  | 60.4  | 69.7  | 69.5  | 78.5  | 79.0  | 79.6  | 75.2  | 73.9  | 63.3  | 48.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 37.4  | 38.7  | 42.7  | 47.5  | 53.0  | 59.7  | 61.2  | 58.7  | 55.7  | 49.7  | 43.2  | 37.3  |  |  |
|                                                                        |             | MCDB   | 46.6  | 49.6  | 56.3  | 65.0  | 66.8  | 75.5  | 78.1  | 78.5  | 75.4  | 64.7  | 56.2  | 46.0  |  |  |
|                                                                        | 5%          | WB     | 35.7  | 36.0  | 40.3  | 44.0  | 51.2  | 57.3  | 59.5  | 57.4  | 53.9  | 47.1  | 40.7  | 35.1  |  |  |
|                                                                        |             | MCDB   | 44.1  | 45.4  | 52.5  | 58.1  | 64.6  | 72.8  | 77.2  | 77.3  | 73.6  | 60.7  | 51.6  | 44.4  |  |  |
|                                                                        | 10%         | WB     | 34.1  | 33.7  | 38.0  | 41.3  | 49.2  | 55.3  | 58.1  | 56.1  | 52.1  | 44.9  | 38.0  | 32.9  |  |  |
|                                                                        |             | MCDB   | 42.7  | 41.9  | 48.2  | 52.6  | 61.4  | 69.2  | 76.4  | 75.8  | 71.3  | 57.7  | 47.9  | 41.0  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 20.1  | 20.5  | 21.7  | 22.1  | 22.9  | 25.2  | 30.3  | 30.8  | 28.5  | 23.2  | 19.7  | 18.6  |  |  |
|                                                                        |             | MCDBR  | 22.1  | 22.1  | 27.2  | 32.4  | 32.8  | 34.7  | 36.3  | 38.2  | 37.9  | 32.8  | 24.2  | 19.5  |  |  |
|                                                                        |             | MCWBR  | 12.9  | 12.8  | 15.0  | 17.1  | 16.7  | 16.4  | 14.0  | 14.9  | 16.0  | 16.0  | 13.5  | 12.1  |  |  |
|                                                                        | 5% WB       | MCDBR  | 21.0  | 21.4  | 26.7  | 31.4  | 28.8  | 31.4  | 32.5  | 32.5  | 33.7  | 29.7  | 24.4  | 18.3  |  |  |
|                                                                        |             | MCWBR  | 13.7  | 14.0  | 14.9  | 16.8  | 15.2  | 15.7  | 14.9  | 13.8  | 15.3  | 15.4  | 13.8  | 11.7  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.239 | 0.246 | 0.261 | 0.312 | 0.313 | 0.318 | 0.323 | 0.349 | 0.304 | 0.279 | 0.250 | 0.237 |  |  |
|                                                                        | taud        |        | 2.486 | 2.480 | 2.543 | 2.404 | 2.478 | 2.500 | 2.509 | 2.384 | 2.502 | 2.564 | 2.548 | 2.498 |  |  |
|                                                                        | Ebn at Noon |        | 282   | 302   | 311   | 300   | 302   | 300   | 297   | 284   | 291   | 285   | 275   | 270   |  |  |
|                                                                        | Edh at Noon |        | 20    | 24    | 27    | 35    | 33    | 33    | 32    | 35    | 28    | 23    | 18    | 17    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 489   | 806   | 1180  | 1521  | 1785  | 2081  | 2267  | 1898  | 1387  | 877   | 527   | 380   |  |  |
|                                                                        | RadStd      |        | 37    | 55    | 84    | 87    | 165   | 144   | 113   | 105   | 109   | 72    | 44    | 28    |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (9 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | -0.99 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page